

Manuscript Title

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✉ — Correspondence possible via [GitHub Issues](#)

Abstract

Background & Summary

We scanned a *lot* of different mouse strains and so-called hybrids, both male and female. One scan for each strain, each hybrid and each sex gives already 90 samples.

Methods

Sample preparation

- Mice were euthanized and decapitated
- Mouse skulls were stored in EMA
- Separated into 4 batches of approximately 110 animals, labeled as B `batch_number` `animal number` A and B `batch_number` `animal number` B.

Tomographic imaging

We report the scan settings and reconstruction parameter data as suggested by¹. All logfiles of all acquired scans for this project are directly available online at <https://github.unibe.ch/david-haberthuer/InMice/tree/main/logfiles>.

All scans were acquired on a SkyScan 1272 (Bruker microCT, Kontich, Belgium), equipped with an automatic sample changer. The machine is equipped with a Hamamatsu L11871_20 x-ray source, and a XIMEA xiRAY16 x-ray camera. Acquisition was performed in cone-beam mode, inherent to the machine, with a source-to-detector distance of 174.1 mm and a source-to-sample distance of 100.0, 94.1 or 78.9 mm.

QA

- Use a collection of logfile wrangling code² to go through *all* the log files of all the acquired scans
 - Use this to surface issues related to acquisition (wrong setting) and reconstruction
- Look at average and maximal brightness of (a subset of) all the projection images acquired
 - Use this to surface issues related to acquisition, e.g. sometimes the x-ray source inadvertently shut down, or the counts were too low on the camera, etc.

Image processing

- Jupyter³ notebooks, available here: <https://github.unibe.ch/david-haberthuer/InMice/>, for reproducible research.
 - Ingest complete, uncropped reconstructions with `dask`^{4/}
 - Crop, based on axial MIPs
 - Save cropped data out as `.zarr`-files, ready to be loaded with `n5-ij`⁵ in Fiji⁶
 - Save in other formats, to either use 3D Slicer^{7/8} or Dragonfly^{9/}

Data Records

Technical Validation

Usage Notes

Code Availability

- [Jupyter notebooks](#)

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Author Contributions

[Contributor Roles Taxonomy \(CRediT\)](#), as defined in¹⁰:

- [Conceptualization](#): Bernhard Voelkl, Hanno Würbel
- [Data curation](#): David Haberthür, Larisa Petra Kaija
- [Formal analysis](#): David Haberthür, Bernhard Voelkl
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Competing Interests

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David Haberthür	None	2025-06-27
Pui Ching Chu	None	2025-06-27
Larisa Petra Kaija	None	
Bernhard Voelkl	None	2025-08-19
Hanno Würbel	None	2025-08-19

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